

## MTH441: Lab 2

### Questions

1. Consider the `cars` dataset in base `R` which contains information on the distance a car traveled after breaking. Fit a simple linear regression model and view the summary of the model using the following commands

```
# Fit linear regression model
# automatically fits with intercept
fit <- lm(dist ~ speed, data = cars)
# view summary
summary(fit)
```

The `summary` function reports the following:

- a. a `summary` of the residuals
- b. table of OLS coefficients, with estimate, standard error,  $t$ -value and  $p$ -value
- c. residual standard error with degrees of freedom
- d.  $R^2$  and adjusted  $R^2$
- e.  $F$ -statistic for model fit, with degrees of freedom and  $p$ -value

Verify all of the information obtained in the summary table using the formulas and methods you have learned in class.

2. Consider the `Boston` dataset in the `MASS` and using `?Boston` learn about the data. The `medv` variable is the response.
  - a. Fit a multiple linear regression model, with intercept.
  - b. Is the model useful?
  - c. Which factors lead to lower median value of the homes?
  - d. Which factors lead to higher median value of the homes?
  - e. Which factor seems to most significantly affect the value of the home?
  - f. Which factor seems to most greatly affect the value of the home? How is this question different from the previous one? (Note: there are two ways to answer this question.)
3. For the `Boston` dataset:
  - a. Which is the least useful covariate in the model?

- b. Fit a simple linear regression model (with intercept) with `medv` as the response and the least useful covariate. Is the covariate still useless?
- c. Can you explain what happened here?

4. **Multiple comparisons and Bonferonni Correction:** As discussed in class, the `lm` summary table produces marginal tests for significance of individual regression coefficients. Let us understand how this can be not great. Let us first generate data from a model where we know the null hypothesis  $H_0 : \tilde{\beta} = 0$  is true.

```
n <- 1e3
p <- 99
# true beta is 0
beta <- rep(0, p+1)

# Generate arbitrary X
X <- matrix(rnorm(n*p), ncol = p, nrow = n)
X <- cbind(1, X)

# Generate y: response with variance = 1
y <- rnorm(n, mean = X %*% beta, sd = 1)
```

Do the following

- a. Calculate the OLS estimator
- b. For each of the 100 components of the OLS estimator, perform a  $t$ -test with size 0.05.
- c. Obtain the proportion of coefficients that reject the null hypothesis and compare this to 0.05.
- d. Why do you observe what you observe?
- e. Repeat the experiment 100 times and track in how many datasets were all  $\beta$ s correctly declared to be the zero vector?
- f. Thus, to check whether the whole vector is the zero vector or not, is this test effective or not?

5. **Multiple comparisons and Bonferonni Correction:** In the previous question, instead of using quantile  $t_{\alpha/2, n-p-1}$  use  $t_{\alpha/(2(p+1)), n-p-1}$ . This is the Bonferroni correction. For the same data generation mechanism, do the following

- a. Calculate the OLS estimator
- b. For each of the 100 components of the OLS estimator, perform a  $t$ -test with the above  $t$ -quantile.
- c. Obtain the proportion of coefficients that reject the null hypothesis and compare this to 0.05.
- d. Why do you observe what you observe?
- e. Repeat the experiment 100 times and track in how many datasets were all  $\beta$ s correctly declared to be the zero vector?
- f. Can you explain the difference from the previous implementation and why do you see what you see?